Calculation 1: Direction field & Equilibrium solution

In the previous section you have derived the differential equation

$$\frac{dP}{dt} = 0.7P(t) - 20,$$

still with the initial condition $P(0) = 30.$

Can you estimate the number of rainbowfish $P(t)$ *over time?*

To answer this question, we will first construct a *direction field*, which is a special type of graph. In the next video, you will see how you can do this. In the video also some solutions are estimated. See for yourself!

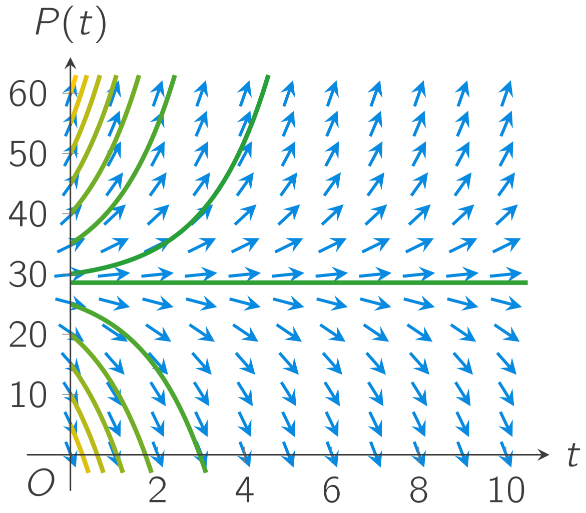
Direction fields

5:51 / 5:51 2.0x HD

Video Subtitles Other

Subtitles (captions) in other languages than provided can be viewed at YouTube. Select your language in the CC-button of YouTube.

In the video you have seen how you can construct the *direction field* of our differential equation and how you can sketch solutions to the differential equation if you know the initial condition:



Exercise A

6/6 points (graded)

Keyboard Help

The direction field above contains several sketches of solutions. As you might have seen, some of the sketched solutions show an increase in the number of rainbowfish. Other solutions decrease. The behaviour depends on the initial condition. Drag and drop the following initial conditions to the correct category.

Increasing number of rainbowfish	Decreasing number of rainbowfish

Submit You have used 1 of 3 attempts.

Reset Show Answer

FEEDBACK

✔ Correctly placed 6 items.

✔ Good work! You correctly categorised the initial conditions and their corresponding solutions using the direction field.

✔ Your highest score is 6.0

A solution $P(t) = P_e$ which neither increases nor decreases, is an *equilibrium solution* of the differential equation. Because an equilibrium solution does not change, it has the property

$$\frac{dP}{dt} = 0.$$

In other words, the equilibrium solution $P(t) = P_e$ is constant in time. Because it is constant, other words for an equilibrium solution are an *equilibrium point* and a *stationary point*.

Exercise B

1/1 point (graded)

Our differential equation is

$$\frac{dP}{dt} = 0.7P(t) - 20.$$

What is the exact value of the equilibrium solution P_e?

Give your answer as a fraction.

200/7

✔ **Answer:** 20/0.7

$\frac{200}{7}$

Answer

Correct: Indeed, you need to divide 20 by 0.7.

Submit You have used 1 of 3 attempts

Answers are displayed within the problem

Now you can sketch a direction field for a differential equation, and calculate its equilibrium solutions. Next, you will investigate the equilibrium solutions further.